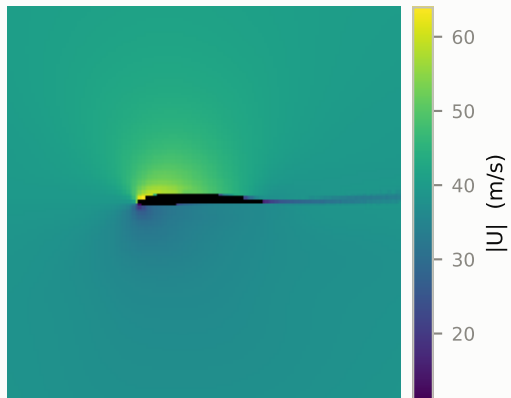


Self-auditing neural CFD: the surrogate checks its own prediction against the governing physics — and decides

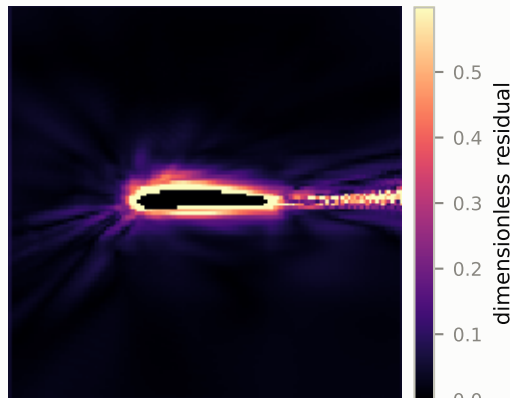
1 · Predict

surrogate speed field



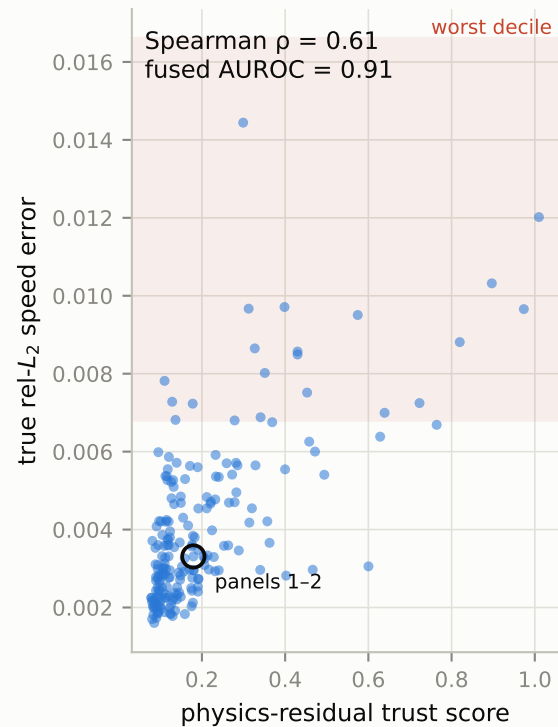
2 · Audit

physics-residual map (no ground truth)



3 · Calibrate

trust score ranks true error (200 cases)



4 · Decide

reject the least-trusted $\rightarrow$ near-oracle

